

**SAVEETHA SCHOOL OF ENGINEERING**  
**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES**

**INSTITUTE OF AI & ML**

**DEPARTMENT OF COMPUTATIONAL INTELLIGENCE**

**CSA14 - COMPILER DESIGN**

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**DEPT:- AI & ML**

1. The Company ABC runs with employees with several departments. The Organization manager had all the mobile numbers of employees. Assume that you are the manager and need to verify the valid mobile numbers because there may be some invalid numbers present. Implement a LEX program to check whether the mobile number is valid or not.

```
[boddayamini@BODDAs-MacBook-Air ~ % brew install flex
Warning: flex 2.6.4_2 is already installed and up-to-date.
To reinstall 2.6.4_2, run:
  brew reinstall flex
[boddayamini@BODDAs-MacBook-Air ~ % mkdir lex_mobile && cd lex_mobile
[boddayamini@BODDAs-MacBook-Air lex_mobile % nano mobile.l
[boddayamini@BODDAs-MacBook-Air lex_mobile % flex mobile.l
[boddayamini@BODDAs-MacBook-Air lex_mobile % gcc lex.yy.c -o mobile
[boddayamini@BODDAs-MacBook-Air lex_mobile % ./mobile
Enter a mobile number to verify: 6385682789
Valid Mobile Number: 6385682789

2354619
Invalid Mobile Number (Incorrect format/length): 2354619
```

```
~/lex_mobile - nano mobile.l
File: mobile.l

%{
#include <stdio.h>
%}

%%
[6-9][0-9]{9} { printf("Valid Mobile Number: %s\n", yytext); }
[0-9]+ { printf("Invalid Mobile Number (Incorrect format/length): %s\n", yytext); }
[ \t\n]+ ; /* Ignore whitespaces and newlines */
. { printf("Invalid Input (Contains non-digit characters): %s\n", yytext); }

%%

int yywrap() {
    return 1;
}

int main() {
    printf("Enter a mobile number to verify: ");
    yylex();
    return 0;
}
```

2. Write a LEX Program to check the email address is valid or not.

```
[boddayamini@BODDAs-MacBook-Air lex_mobile % nano email.1
[boddayamini@BODDAs-MacBook-Air lex_mobile % nano email.1
[boddayamini@BODDAs-MacBook-Air lex_mobile % ./email_check
Enter an email address to verify: yamini@gmail.com
Valid Email Address: yamini@gmail.com
yamini@gmailcom
Invalid Email Address: y
Invalid Email Address: a
Invalid Email Address: m
Invalid Email Address: i
Invalid Email Address: n
Invalid Email Address: i
Invalid Email Address: @
Invalid Email Address: g
Invalid Email Address: m
Invalid Email Address: a
Invalid Email Address: i
Invalid Email Address: l
Invalid Email Address: c
Invalid Email Address: o
Invalid Email Address: m
```

```
~/lex_mobile -- -zsh                                ~/lex_mobile -- nano email.1
File: email.1

%{
#include <stdio.h>
%}

%%
[a-zA-Z0-9-_%+]+@[a-zA-Z0-9-]+\.[a-zA-Z]{2,}\n { printf("Valid Email Address\n"); }
[^\n]+\n { printf("Invalid Email Address\n"); }
[ \t]+ ; /* Ignore spaces and tabs */

%%

int yywrap() {
    return 1;
}

int main() {
    printf("Enter an email address to verify: ");
    yylex();
    return 0;
}
```

Get Help   WriteOut   Read File   Prev Pg   Cut Text   Cur Pos  
Exit   Justify   Where is   Next Pg   UnCut Text   To Spell

3. Write a LEX program to check whether the given input is a digit or not.

```
UW PICO 6.09 File: digital.l
%{
#include <stdio.h>
%}

%%
[0-9] { printf("It is a digit\n"); }
. { printf("It is not a digit\n"); }
%%

int main()
{
    yylex();
    return 0;
}

int yywrap()
{
    return 1;
}
```

```
[boddayamini@BODDAs-MacBook-Air ~ % nano digital.l
[boddayamini@BODDAs-MacBook-Air ~ % flex digital.l
[boddayamini@BODDAs-MacBook-Air ~ % gcc lex.yy.c -o digit
[boddayamini@BODDAs-MacBook-Air ~ % ./digit
7
It is a digit

a
It is not a digit

^C
```